

MassMarsh
Standard Operating Procedure
Salt Marsh Accretion and Elevation Monitoring

1.0 Purpose and Applicability

Coastal salt marshes exist within a narrow range of physical settings limited to sheltered coastal environments with adequate sediment supply and vertically bounded by the local tidal frame. Thus, the persistence of coastal wetlands relies on their ability to accrete sediment vertically and maintain their elevation relative to sea level through a combination of biophysical processes closely related to tidal hydrology. Because of the fundamental link between marsh surface elevation and local sea level, measuring and monitoring both marsh surface elevation and vertical accretion are important metrics for evaluating current marsh status, defining trajectories, and monitoring the efficacy of marsh restoration projects. This document addresses both vertical accretion and surface elevation (and elevation change) as indicators of marsh condition and processes.

Vertical accretion refers to vertical growth of the salt marsh surface due to the deposition of suspended particles (sediment) during inundation and the accumulation of plant material, including both below ground root material and the partially decomposed litter from plants growing on the marsh surface (Schuerch, 2012). **Elevation change** refers to vertical changes in the height of the wetland surface relative to a fixed vertical benchmark. Vertical accretion contributes to elevation changes, though the two are not necessarily equivalent due to subsurface compaction that may result from dewatering of pore spaces and/or decomposition and oxidation of organic matter within the soil underlying the marsh surface. Marsh surface elevation change is subsequently equal to accretion minus any shallow subsurface compaction that occurs combined with any deeper geologic subsidence (e.g., tectonic or isostatic). In terms of assessing salt marsh vulnerability to sea level rise, the rate of marsh surface elevation change relative to sea level rise is the critical factor and direct measures of elevation change are most relevant. In practice however, vertical accretion is more easily and more commonly measured. While not a substitute for measurement of elevation change, the greater ease of measuring accretion rates makes it suitable for more widespread application. Because rates of vertical accretion provide a maximal estimate of elevation change in the absence of any shallow subsurface compaction, evaluation of accretion rates is suitable for assessing the potential for salt marsh surfaces to keep pace with sea level rise.

2.0 Availability, Suitability, and Applications of Accretion and Elevation Data & Methods

2.1 The Surface elevation table–marker horizon (SET-MH) method provides millimeter resolution measurement of wetland elevation and accretion, and is the global standard for assessing coastal salt marsh resilience to sea level rise (Webb et al., 2013) (Figure 1). The SET-MH method is widely used, with well-established protocols. A comprehensive description of protocols surrounding installation, monitoring, and data processing of SET-MH stations is provided by Lynch et al. (2015) and this should be considered the primary guidance for standard operating procedures related to SET-MH installations. A 2024 special edition of the journal *Estuaries and Coasts* focused on recent advances in coastal wetland elevation dynamics and reviews applications and advances in the SET-MH method (Cahoon, 2024; Cahoon and Guntenspergen, 2024). A brief review of SET-MH can be accessed at [this USGS web page](#).

In this document, we focus specifically on considerations relevant to the *application* of SET-MH stations to the objectives of the MassMarsh monitoring program. Perhaps the single most significant limitation

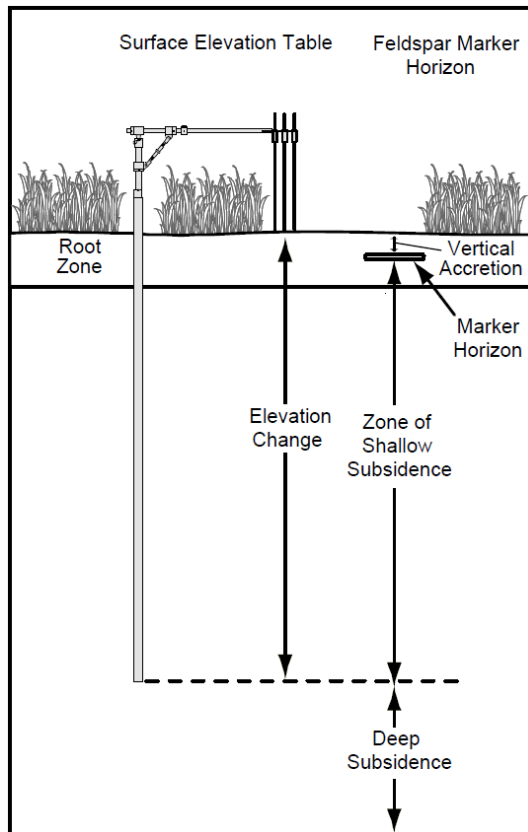


Figure 1 - Conceptual diagram of a Surface Elevation table (SET) showing portions of the soil profile measured by the SET and soil marker horizon. From Cahoon et al. (2011).

of SET-MH monitoring is the limited spatial resolution, tracking elevation dynamics across approximately 1.5 m² area of the wetland surface. A growing body of literature has explored applications of extending the spatial coverage of SET observations using terrestrial lidar or precision GPS survey methods (Cahoon and Guntenspergen, 2024), though it generally remains the case that caution is urged when SET measurement trends are applied over larger areas. A common limitation of analyses of relative sea level rise influencing specific wetland areas is an assumption that deep subsidence at wetland study sites is the same as that occurring at the locations of tide gauge records. Recent studies utilizing SET data have highlighted substantial differences in the rate of deep subsidence beneath wetlands and regional tide gauge sites (Doar and Luciano, 2023; Hensel et al., 2024), emphasizing the fact that the most reliable estimates of relative sea level rise for individual wetlands necessitate direct measurement of vertical land motion below both wetland study sites and tide gauge stations (Cahoon, 2015).

Surface elevation tables are typically paired with marker horizons to provide comprehensive insight into wetland elevation dynamics. The SET-MH method is best suited to long-term (decadal scale or longer) monitoring of marsh health, with specific regard to wetland surface

elevation relative to sea level. Numerous existing SET-MH installations are sited throughout Massachusetts wetlands, primarily in locations managed by the U.S. Fish and Wildlife Service (e.g. Parker River and Monomoy National Wildlife Refuges), the U.S. National Park Service (e.g. Cape Cod National Seashore), and the National Estuarine Research Reserve System (Waquoit Bay). Because of the requirements for permanent installations and ongoing, annual observations collected in a consistent and precise manner, SET-MH installations must be sited in locations where continued access is ensured and where the continued availability of suitably trained observers is expected. Inexpensive and easily deployed un-paired marker horizons can supplement existing long-term SET-MH installations allowing for increased geographic coverage of Massachusetts salt marshes and increased spatial coverage of surface accretion within individual wetlands, though they do not provide information about elevation change. Un-paired MH installations should follow protocols described by Lynch et al. (2015), and described briefly below. While SET-MH and un-paired MH installations are best suited to long-term monitoring, they may be appropriate to evaluating the efficacy of some restoration projects, especially those that affect tidal hydrology and the delivery of sediment to the wetland surface, ultimately influencing accretion processes. Though the application of SET-MH or MH observations to monitoring restoration efficacy requires at least several years of observations prior to and following project implementation to be able to detect a change in the rate of marsh accretion or elevation change.

2.1a Brief description of marker horizon installation

Detailed protocols for MH installation can be found in Lynch et al. (2015). Here we report a quick start guide to help prepare for field application. Lynch et al. (2015) should be reviewed before actual installation.

Materials:

- Roughly 10 lb feldspar clay or sand per MH
 - Often good to experiment with both as persistence will vary by site depending on bioturbating organisms present
 - G-200 feldspar clay is ideal as it is bright white and often available from pottery supply vendors
- PVC stakes to mark MH plot corners; ½" diameter, 2-3 foot length
- 50 cm by 50 cm frame
- N95 respirator if using feldspar clay to avoid inhalation

Installation steps:

1. Lay frame down on a uniform section of marsh platform large enough to contain the frame
2. Use a cup or sifter to evenly sprinkle no more than 1 cm material within the frame area
3. Mark corners of MH with PVC stakes

Making a MH measurement

1. After approximately one year, revisit the MH location.
2. Use a serrated knife to extract a cone shaped core approximately 5 cm in depth and diameter
 - a. Focus on the North edge of the MH in yr 1, the east edge in yr 2, the south edge in yr 3, and the west edge in yr 4.
 - b. MH often have to be reestablished after 1-4 years as bioturbation (and or erosion) obscure the MH layer over time.
3. Use calipers to measure the distance from the land surface to the MH at 3-5 locations as resources permit.

2.2 High precision survey methods (RTK GPS or total station) are appropriate for monitoring elevation components of wetland restoration projects such as thin layer placement, construction of marsh habitat mounds or marsh islands, and ditch remediation. To do so, surveys should occur prior to any interventions, immediately after construction or implementation, and one, three, and five years after construction. When re-occupying a transect will be necessary to achieve monitoring goals, transect start and end points should be monumented with semi-permanent markers (e.g., PVC pipes or rebar). Precision survey techniques are most useful for detecting changes in elevation of several centimeters or more, with limitations constrained by the typical 2 cm vertical accuracy of RTK GPS measurements and the challenges of accurately obtaining repeat measurements of an uneven and vegetated marsh surface. These factors limit applications of survey techniques for monitoring the efficacy of passive sediment augmentation projects or long-term monitoring of marsh surface elevation change. Both the use of RTK GPS or total stations for surveys requires that users are trained in their operation and familiar with standard quality assurance and quality control procedures used in land surveying. A coordinate reference system with units of measurement and horizontal and vertical datums must be identified. MassMarsh recommends recording all geospatial survey data relative to the North American Datum 1983 (NAD83) using the Massachusetts State Plane Coordinate System, Mainland (Fipzone 2001) and the North American Vertical Datum 1988 (NAVD88) using meters.

Use of a total station requires that survey points are measured relative to a fixed benchmark of precisely known elevation relative to NAVD88.

A goal of elevation surveys in monitoring the efficacy of restoration efforts includes evaluating whether sediment added to a marsh surface remains in place or that remediated ditches in fill with sediment. Elevation surveys should thus be designed to capture variations at scales appropriate to individual projects. Monitoring of each constructed feature or remediated feature should involve multiple transects oriented as described below. Survey points on each transect should be spaced at 1 m intervals for evaluation of large (>5m diameter) habitat mounds and thin layer placement applications, with additional points added at any notable changes in surface slope. For surveys of smaller features, such as habitat mounds or individual ditches, survey point spacing should be 0.2 m, also with additional points added at notable changes in surface slope. For evaluation of small (<5m diameter) marsh habitat mounds or constructed marsh islands (based on assumed constructed geometry with diameter less than ~30 m), two survey transects should be established in orthogonal orientations, intersecting at or near the center of a feature. For circular features, it is generally recommended that transects be oriented in both north-south and east-west directions (Figure 2). For oblong features, it may be more appropriate to orient a primary transect along the centerline of the long axis of the feature, with a second transect perpendicular to the long axis at its midpoint. Survey transects should extend beyond the perimeter of the constructed feature in each direction, by a distance equal to or exceeding one half the diameter of the constructed feature.

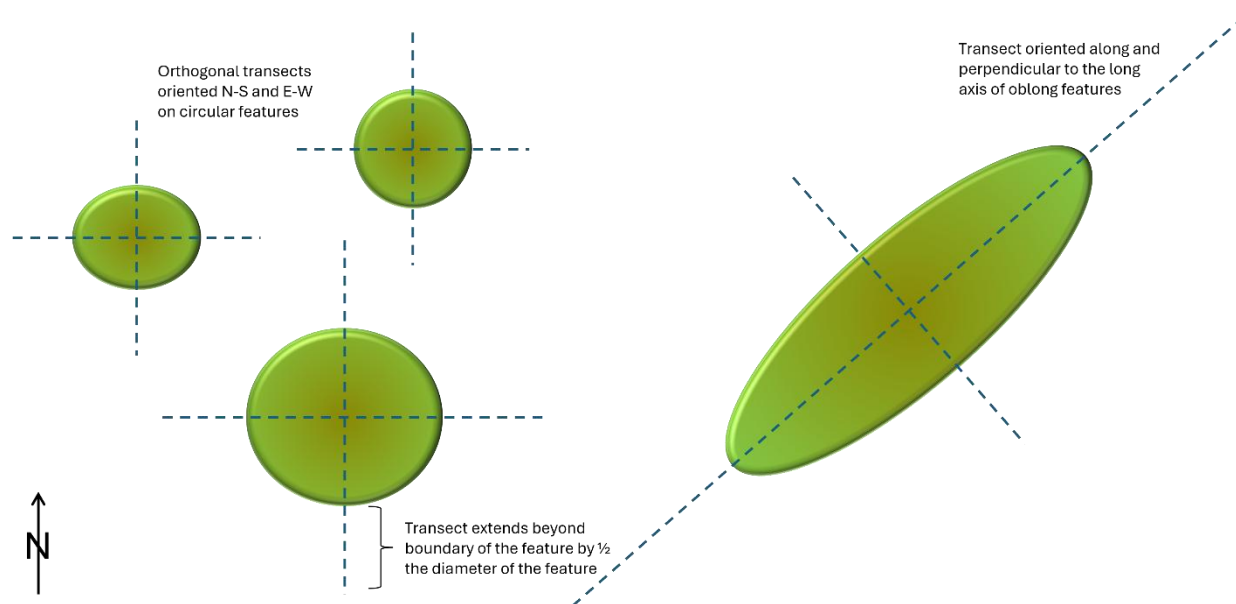


Figure 2 Recommended layout of transects relative to constructed marsh habitat mounds (marsh islands) as indicated by the dashed lines.

For tracking the elevation of larger interventions (Figure 3), survey layout should include identification of a primary transect aligned along the long axis of the region of interest. Three orthogonal transects should be defined relative to the primary transect, situated at 20%, 50% and 80% of the distance along the length of the primary transect. The width of the region of interest should be defined by the average length of the feature as measured along these three orthogonal transects. Finally, an additional two transects should be defined parallel to the primary transect and spaced at 20% and 80% width of the

feature. Each survey transect should extend beyond each edge of the feature being evaluated by a minimum distance of at least 10% of the feature length as measured along that transect.

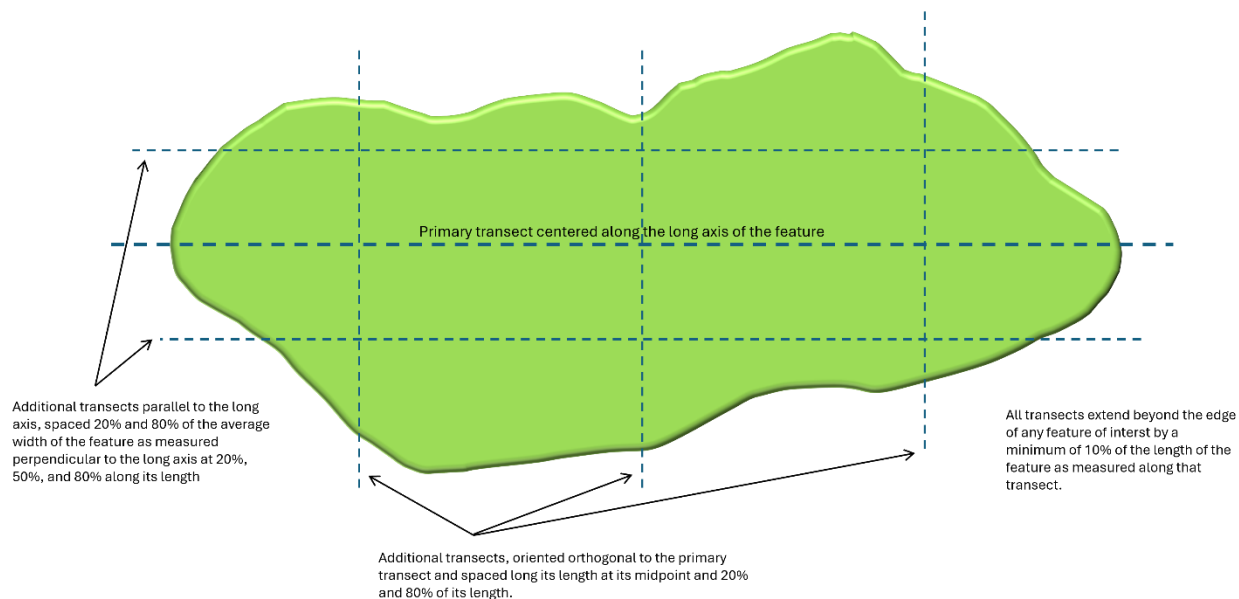


Figure 3 Proposed survey layout for regions of interest >50 m in length.

To monitor the efficacy of ditch remediation (see glossary), the elevation of the placed thatch should be measured relative to a fixed vertical datum. The intent is to measure changes in the elevation of the revegetating, restored ditch bottom as a result of intervention. Three ditch-perpendicular transects should be established, with observations spaced at 0.2 m intervals. The transects should be spaced at 20%, 50%, and 80% of the total length of the ditch (Figure 4). Transects should extend a minimum of three ditch widths on either side of the ditch being evaluated. A ditch-parallel transect should also be collected, with elevations captured every 5 m along the length of the remediated ditch. Two elevations should be recorded at each observation point on the ditch-parallel transect, the top of the thatch, and the bottom of the pre-restoration ditch via probing with a survey rod.

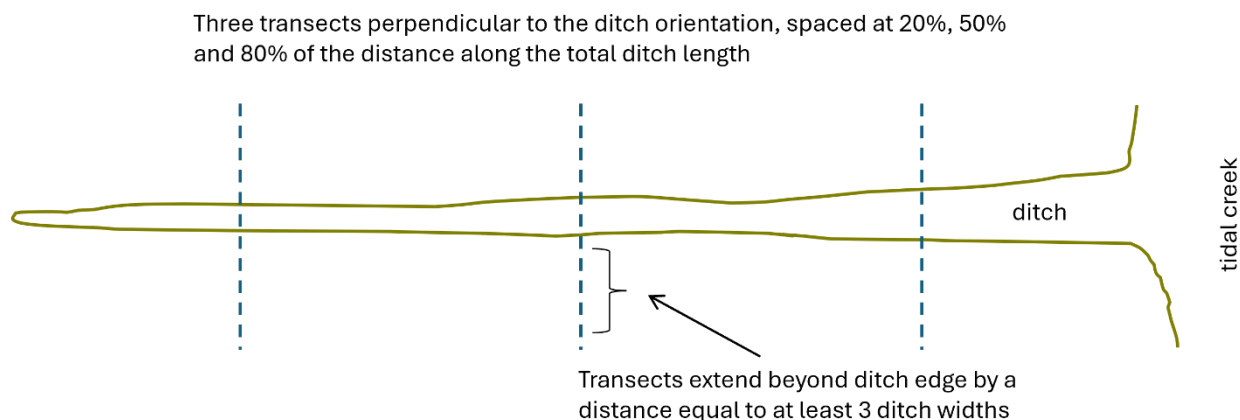


Figure 4 Proposed survey layout for monitoring remediation of ditches

3.0 Evaluation of Marsh Status using Regional Geospatial Datasets

Several existing regional and national level geospatial datasets offer relevant metrics of marsh surface elevation, elevation change, and rates of accretion. Brief descriptions, links to source data, and comments on possible applications and limitations of these datasets are included below.

3.1 Elevation of marsh units in Massachusetts salt marshes.

This dataset, released by the USGS and developed in collaboration with the Massachusetts office of Coastal Zone Management provides the lidar based elevation distribution of salt marshes in Massachusetts in terms of the mean elevation of conceptual marsh units (Ackerman et al., 2021). Conceptual marsh units are subdivisions of wetland complexes defined by ridgelines separating surface drainage divides. Small drainage areas were iteratively merged with their nearest larger neighbor to produce marsh units with a minimum area of 5,000 square meters. The mean elevation for each marsh unit was determined based primarily on the 1-meter resolution Coastal National Elevation Database (CoNED), as well as a 1-meter resolution USGS DEM, and the 1/9- arc-second resolution USGS National Elevation Dataset (NED). Mean elevation of marsh units provides a means of assessing the potential vulnerability of marsh platforms to drowning by sea level rise. These data currently present a single snapshot in time of the marsh surface elevation, rather than any indication of whether a marsh unit may be gaining or losing elevation relative to current rates of sea level rise. Furthermore, stable marsh surface elevation may exist at different elevation given variations in local tide range. While low elevation areas of marsh are likely to be more vulnerable to flooding, these data should be used in combination with other observations to assess likely trajectories of marshes. It is also important to note that the spatial scale of conceptual marsh units may obscure areas within larger marsh units that are losing elevation relative to sea level and which could potentially benefit from some form of intervention. In summary, marsh units characterized by especially low mean elevation may be more prone to future drowning. However, the actual vulnerability of these areas should be evaluated in combination with other data sources describing the biophysical characteristics of the marsh at higher spatial resolution and assessment should also consider trends in marsh properties over time. Marsh Unit Elevation data can be explored online through the [USGS Massachusetts Coastal Wetland Synthesis Web App](#)

3.2 Relative tidal marsh elevation, Z^*_{MHW}

Z^* is a physical metric that normalizes elevation to tidal amplitude at mean high water (MHW), available at 30×30 m resolution spanning the conterminous USA in 2010 (Holmquist and Windham-Myers, 2021; 2022).

3.3 Lidar digital elevation data

Airborne light detection and ranging, or Lidar, elevation data provide spatially continuous, high resolution (typically 1m or better horizontal resolution) coverage of coastal elevation, with widespread application beginning in the late 1990s (National Oceanic and Atmospheric Administration, 2012). Stated vertical accuracy is typically <0.2 m and for most recent data <0.04 m. However, dense marsh vegetation can prevent signal returns from the bare earth resulting in a systematic bias, with lidar-derived elevations 0.1 – 0.2 m above surveyed elevations depending on vegetation height and seasonality (Hladik and Alber, 2012). Because offsets exist across marsh areas, lidar is a suitable method for illustrating areas of subsidence that should be targeted for intervention. Lidar data are most commonly obtained from aircraft via light detection and ranging, though ground-based and UAV-based lidar sensors may be employed on smaller spatial scales with improved resolution. High resolution lidar digital elevation models are extremely valuable for visualizing salt marsh morphology, illustrating legacy impacts from human activities within marshes, and for identifying specific locations within large marsh

units which may be currently degraded or most vulnerable to future impacts. Lidar-based hypsographic measurement of marsh areas may provide a means for identifying regions of degraded marsh or anomalous elevation resulting from human activity (e.g. Masselink and Jones, 2024). Repeat lidar surveys facilitate analyses of lateral shoreline change but are more limited in their ability to discern vertical changes at scales relevant to monitoring of marsh processes. The decimeter scale vertical accuracy of lidar measurements relative to typically slow rates of vertical elevation change in salt marshes (on the order of several mm per year) generally preclude the use of lidar surveys for analysis of trends in wetland elevations on timescales of several years to decades, except in the most extreme cases of wetland change. The NOAA Digital Coast Data Access Viewer (<https://coast.noaa.gov/dataviewer/>) provides an online map interface for identifying and downloading available coastal lidar data.

3.4 Northeast USA Tidal Wetland Accretion and Elevation Change Database

A database of accretion and elevation change rates for coastal marshes of the northeastern United States is a compilation of observations published from 1975 through 2022, with ongoing updates (Cook et al., 2023). This dataset reflects available measurements of accretion and / or elevation change derived from a combination of sediment core analyses, marker horizon studies, and SET monitoring. Individual observations provide point measurements of accretion rates or elevation change at marshes throughout the northeastern US. These data are most suited to making general comparisons among areas with multiple observations to evaluate potential differences in marshes' ability to keep pace with sea level rise. Given the limited spatial coverage and significance of individual point measurements caution is warranted when interpreting these data as an indication of wider patterns throughout a marsh complex – especially in systems with limited available data. More value should be given to elevation change assessed during the recent past via marker horizon or SET relative to sediment core-derived estimates that integrate over longer time periods and may not reflect current conditions. These data are most valuable for defining baseline conditions against which recent or future changes can be assessed. In addition, the extent of available observations highlights gaps which may warrant additional data collection. Accretion and elevation change data can be viewed and downloaded from the [Northeast USA Tidal Wetland Accretion and Elevation Change Viewer](#).

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